BUSINESS ANALYTICS PROJECT

Auto Insurance Claims Fraud Analysis

Submitted by: Tomorrow Lake

Business Problem

Auto insurance companies have a hard problem with claims handling. Special Investigations Units (SIU) catch real fraud, but investigations take time and slow down legitimate payouts. Simple automated rules tend to miss new fraud patterns and flag too many honest claims. The real question is one of prioritization. Given that investigators only have so much capacity, which claims should be sent for review first to catch the most fraud without creating problems for the people filing legitimate claims?

My project looked at 1,000 auto insurance claims to figure out which patterns most strongly predict fraud, and then to recommend a simple routing rule that an SIU could actually use.

Primary Customer

The primary customer is the **Director of Special Investigations (SIU)** at a mid-sized auto insurance carrier. The Director sets the rules for which claims go to investigators, and has to balance two things at once: catching as much fraud as possible, while not making legitimate claimants wait for their payouts.

Other people who would care about this work include the claims operations team (who handle the actual routing), the analytics team (who maintain detection systems), and customer experience leadership (who hear it when customers feel like the company is treating them like suspects).

What I Did

I started with a dataset of 1,000 claims, of which 24.7 percent were labeled as fraudulent. I cleaned it up, profiled it, and then ran a series of analyses in SQL and Tableau to figure out which claim attributes were most associated with fraud. My approach had four steps:

- Look at fraud rates across different claim attributes (severity, collision type, witness count, whether a police report was filed, etc.)
- Pick out the strongest individual signals by comparing fraud rates side by side
- Combine those signals into a simple red-flag count from 0 to 4
- Calculate the trade-off between investigator workload and fraud capture for different routing thresholds

Key Findings

- **Major Damage incidents had a 60.5 percent fraud rate,** compared to 7 to 13 percent for other severity levels. That is roughly six times the population baseline.
- **Red-flag count maps to fraud probability in a clean linear way:** 14.2 percent at 0 flags, 25.2 percent at 1, 33.1 percent at 2, and 41.0 percent at 3 or 4.
- **Recommended routing rule:** any claim with 2 or more red flags goes to SIU. That is 27.5 percent of claim volume but it captures 38.1 percent of fraud, equal to about $5.5 million in flagged fraudulent claims.

Finished Product (Visualization)

The deliverable is an interactive Tableau dashboard built for executives. It has four KPI tiles at the top with the headline numbers, plus three analytical views: a fraud-rate-by-severity bar chart, a red-flag distribution chart, and a top-fraud-profiles ranking. I used a green to red color gradient throughout so the risk levels are easy to read at a glance, and I sized everything to fit on a single screen so the Director would not have to scroll.

Live dashboard: public.tableau.com/app/profile/tomorrow.lake5414/viz/auto_insurance_dashboard/AutoInsuranceClaimsDashboar

Repository (code, data, queries, executive memo): github.com/tomorrowlake3/Auto_Insurance_Claims_Analysis

Companion executive memo: submitted as the second file with this application. It has the full findings, recommendations, and methodology notes.

Skills I Used

- Data cleaning and profiling in SQL and Excel
- Exploratory analysis across multiple dimensions in SQL
- Visualization design for executive audiences in Tableau
- Translating analytical findings into recommendations someone can actually act on
- Quantifying trade-offs (in this case, investigator workload vs. fraud capture)